Insurance Fraud Detection — Technical Report

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1 Executive Summary

This project builds a supervised **Insurance Fraud Detection** model to identify potentially fraudulent claims for investigation prioritization. The economic value is direct: every avoided false negative (missed fraud) reduces claims leakage, helps stabilize loss ratios, and improves pricing integrity.

Operationally, the model is deployed as a decision-support system. It produces a fraud risk score that can be thresholded into a triage queue for investigators.

2 Methodology and Pipeline

2.1 End-to-End Flow (Leakage-Safe)

1. **Data Cleaning**
   - Removes non-informative identifiers.
   - Handles missing indicators (`?`) and imputes selected categorical fields with mode.
   - Creates `days_to_incident` from policy bind date and incident date.
2. **Feature Engineering**
   - Adds `age_group` and `tenure_group` using domain-inspired bins.
3. **Encoding and Scaling**
   - Ordinal encodes `incident_severity`.
   - One-hot encodes categorical columns (drop-first).
   - Scales numeric columns using persisted `StandardScaler` fit on training data only.
4. **Imbalance Handling (Training Only)**
   - SMOTE is applied only on training split / CV folds (never on test).
5. **Model**

- Tuned XGBoost classifier (`best_xgboost_v1.pkl`).

6. **Threshold Tuning**
   - Threshold is selected on training data only to target high fraud recall, then evaluated once on held-out test set.

2.2 Modular Architecture

- `src/insurance_project/data_loader.py`: dataset loading.
- `src/insurance_project/preprocessing.py`: encoding and numeric scaling.
- `src/insurance_project/features.py`: feature engineering (`age_group`, `tenure_group`).
- `src/insurance_project/modeling.py`: training and evaluation utilities.
- `src/insurance_project/paths.py`: robust path resolution.
- `app.py`: Streamlit UI for real-time inference.

3 Performance Assessment

3.1 Dataset and Evaluation Setup

- Test split: 200 rows (20%), stratified, `random_state=42`.
- Test fraud rate: 0.245.
- Primary ranking metric: PR-AUC (Average Precision) = 0.555.

3.2 Core Metrics (Fraud = Positive Class)

Decision Threshold	Precision (Fraud)	Recall (Fraud)	F1 (Fraud)
0.50	0.635	0.816	0.714
0.74	0.625	0.612	0.619

Table 1: Fraud-class metrics at two operating thresholds.

Why recall is prioritized:

- A false negative is a missed fraud claim, causing direct financial leakage and elevated repeat-offender risk.
- High recall supports a loss-prevention posture; investigator workload is managed downstream via queueing, tiered review, and secondary rules.

4 Visual Diagnostics

4.1 Confusion Matrix (Raw)

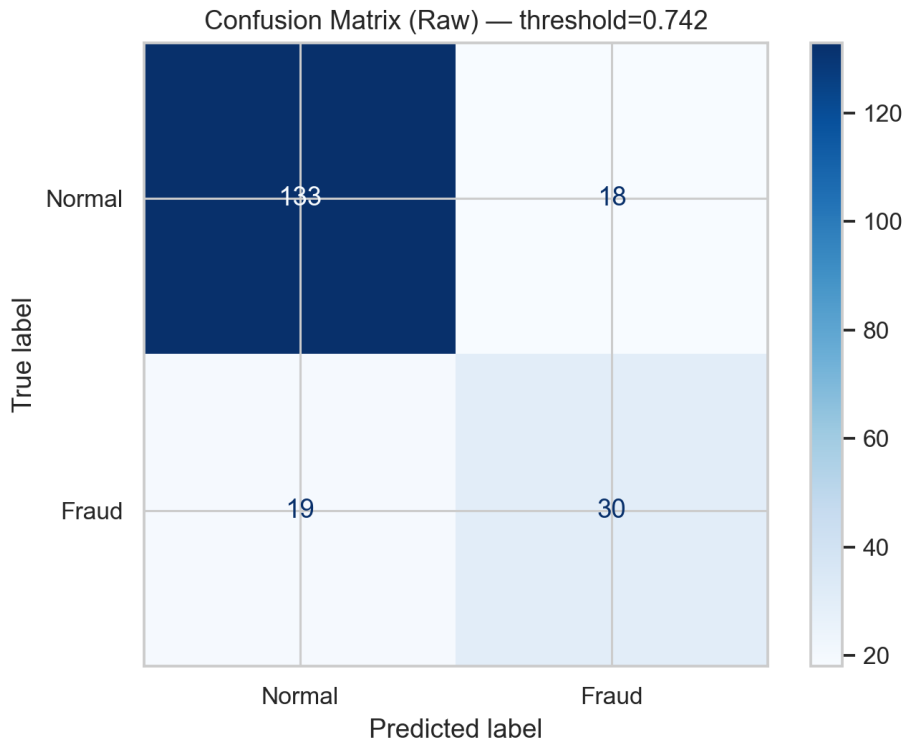

Figure 1: Confusion matrix (raw counts) at threshold 0.74.

This chart shows the absolute number of correct and incorrect classifications at the chosen operating threshold (0.74). It makes the economic trade-off explicit: FN = 19 are missed fraud cases (highest-cost errors), while FP = 18 represent cases sent for investigation that later appear normal.

4.2 Confusion Matrix (Normalized)

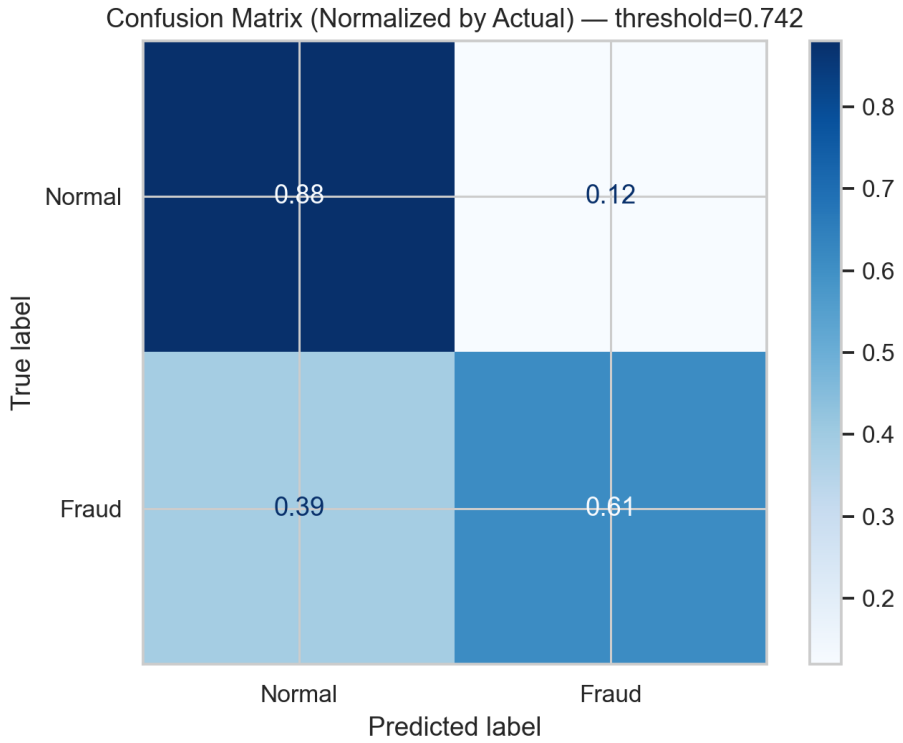

Figure 2: Confusion matrix normalized by true class.

The normalized confusion matrix reports rates per true class, making it easier to compare performance under class imbalance. In particular, fraud-row normalization corresponds closely to fraud recall and is the most relevant view when the business goal is to reduce missed fraud.

4.3 Precision–Recall Curve (PR-AUC)

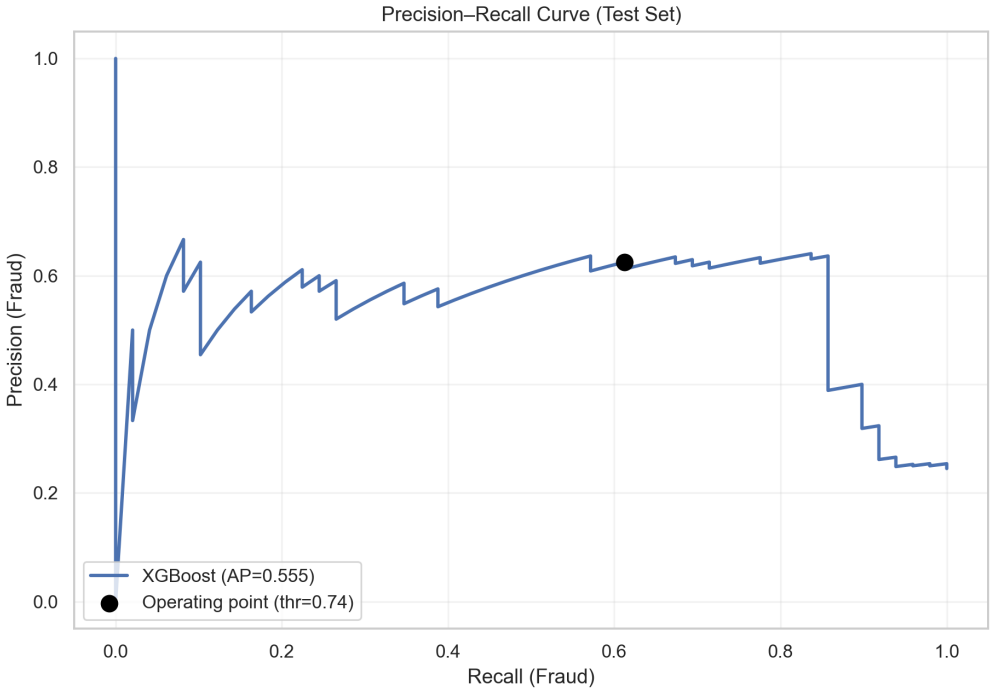

Figure 3: Precision–Recall curve with operating point.

The PR curve characterizes the full precision/recall trade-off across thresholds, and PR-AUC summarizes ranking quality in an imbalanced setting. The marked operating point corresponds to the threshold selected on training data and applied once to the test set, aligning evaluation with a high-recall investigation workflow.

5 Model Interpretability (SHAP Insights)

5.1 SHAP Summary Plot (Top 10)

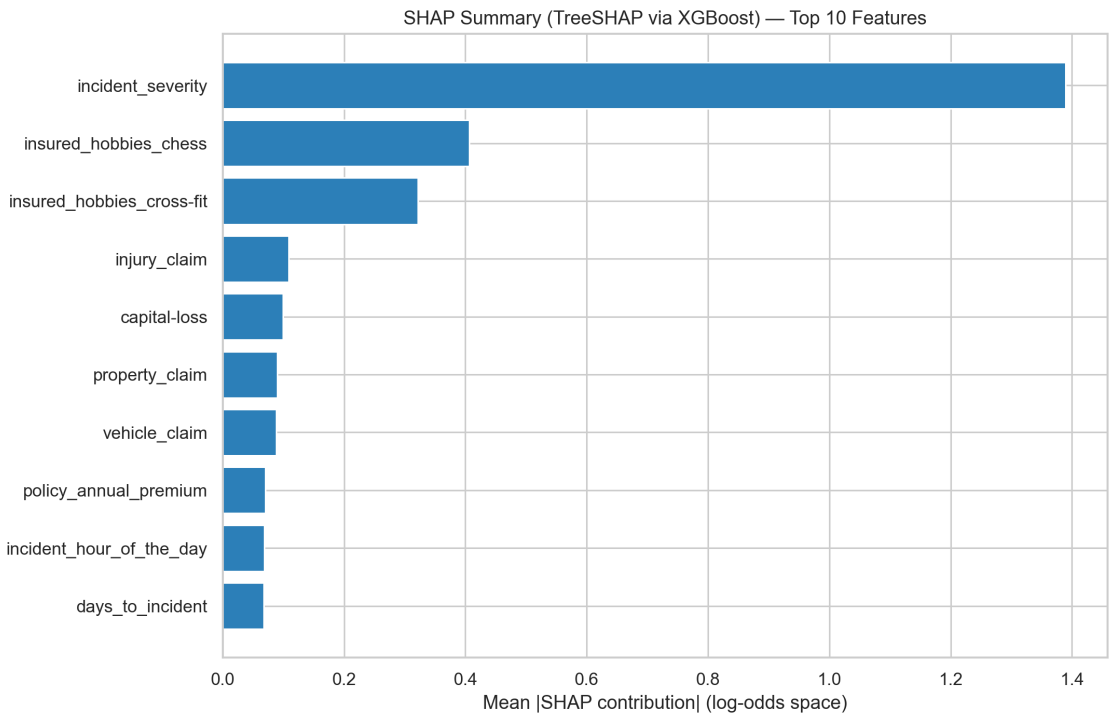

Figure 4: SHAP summary plot of top 10 influential features.

The SHAP summary plot ranks features by average contribution magnitude (mean absolute SHAP) and shows how feature values push predictions toward fraud vs non-fraud. This supports auditable and actionable decision support by surfacing which signals most influence the score.

Feature (model input)	**Mean |SHAP|**
`incident_severity`	1.3894
`insured_hobbies_chess`	0.4065
`insured_hobbies_cross-fit`	0.3221
`injury_claim`	0.1085
`capital-loss`	0.0992
`property_claim`	0.0902
`vehicle_claim`	0.0885
`policy_annual_premium`	0.0699
`incident_hour_of_the_day`	0.0688
`days_to_incident`	0.0674

Table 2: Top 10 features by mean absolute SHAP value.

5.2 SHAP Waterfall (Specific Fraud Instance)

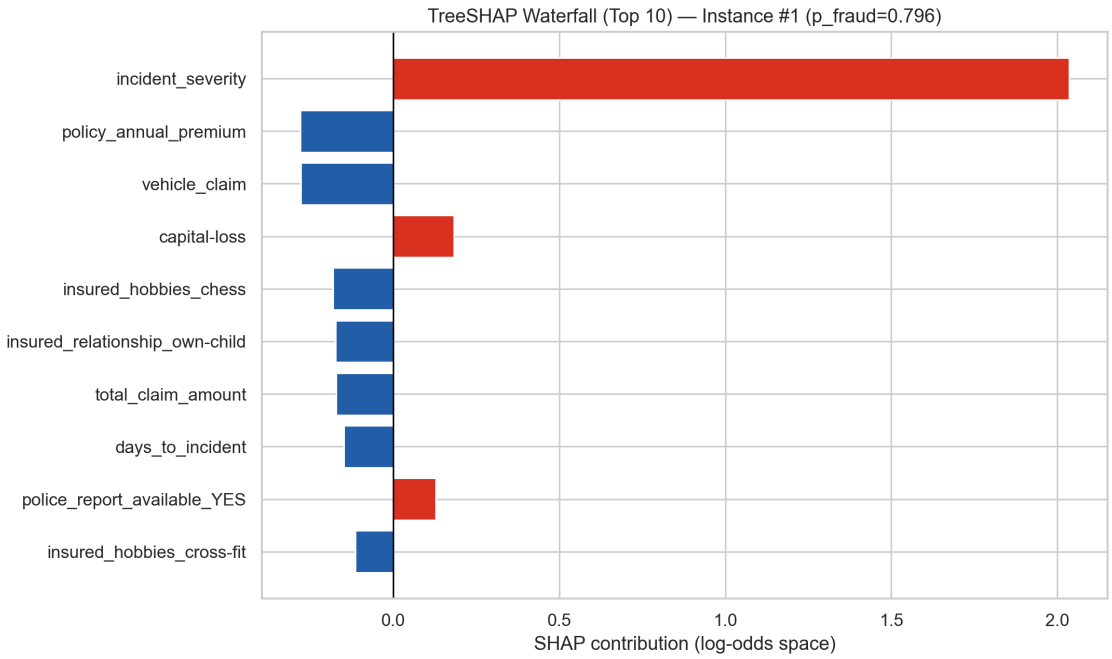

Figure 5: SHAP waterfall explanation for a high-risk claim (test index 1).

This waterfall plot explains a single scored claim (test index 1, predicted fraud probability approximately 0.796) by decomposing the prediction into additive feature contributions. From an operations standpoint, this is suitable for investigator handoff because it highlights the specific red flags driving a high-risk assessment.

5.3 Interpreting Red Flags (Business Logic)

- **Claim severity and amounts** (`total_claim_amount`, `injury_claim`, `vehicle_claim`, `property_claim`): unusually high or inconsistent amounts can indicate inflation or staged losses.
- **Timing anomalies** (`days_to_incident`, `months_as_customer`): very short tenure or incidents soon after binding can indicate opportunistic behavior.
- **Contextual consistency** (`incident_type_*`, `collision_type_*`, `authorities_contacted_*`): reporting patterns can differentiate genuine accidents from fabricated events.
- **Policy and financial signals** (`policy_deductable`, `policy_annual_premium`, `umbrella_limit`, `capital-gains/loss`): coverage structure and financial attributes can correlate with both ability and incentive to commit fraud.

6 Conclusion and Dashboard

This evaluation confirms that the tuned XGBoost model provides meaningful ranking power on an imbalanced fraud dataset (PR-AUC 0.555) and supports a high-recall operating mode to reduce financial leakage.

For production use, the Streamlit dashboard (`app.py`) can serve:

- Real-time scoring for new claims.
- Threshold-driven triage routing.
- Human-in-the-loop review with SHAP-based explanations.

7 Reproducibility Notes (No Leakage)

- Numeric scaling uses persisted `scaler_v1.pkl` fit only on training data.
- SMOTE is a training-only operation and is never applied to the test set or production scoring.
- Threshold selection is performed on training/validation data and then frozen for test/production evaluation.